

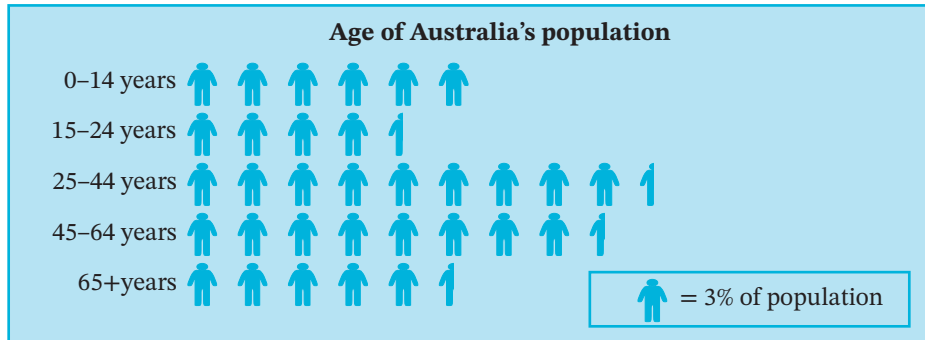


TEST YOURSELF 10

ANSWERS ON P. 601

10.01

- 1 a** What age group has the smallest percentage of the population?

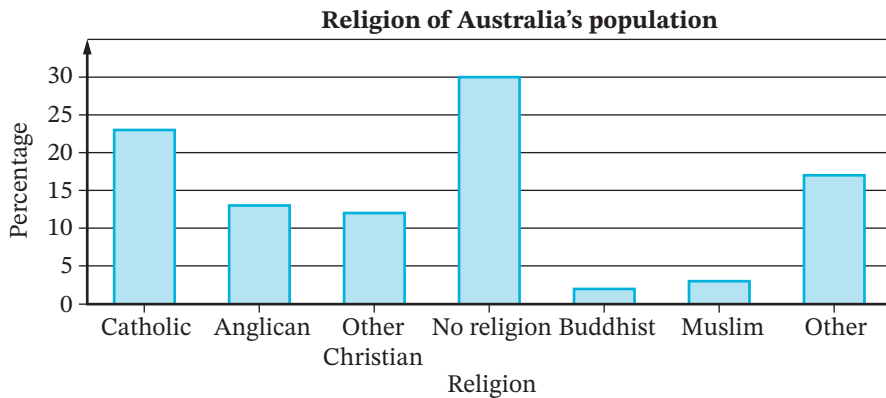


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- b** What percentage are in the 15-24 years age group?
c Why might the 25-44 years group be the largest?

10.01

2

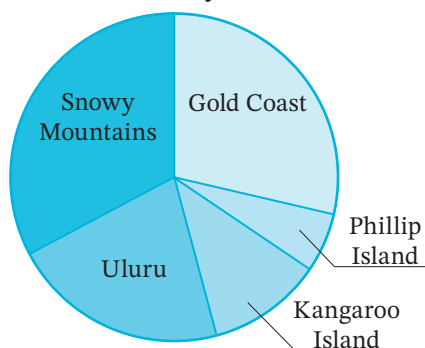


- a** What percentage of the population is Catholic?
b Which religion is followed by 13% of the population?
c How might this graph have been different 20 years ago?

- 3** This sector graph shows the favourite holiday destinations of 80 people surveyed at a city shopping centre.

- What is the most popular destination?
- Which holiday destination was preferred by 17 people?
- Estimate how many people preferred the Gold Coast.
- True or false: Fewer than 10 people preferred Phillip Island.

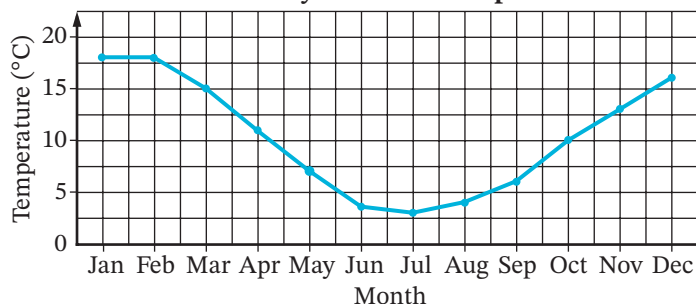
Favourite holiday destinations



10.01

- 4** This graph shows temperature data for Dubbo.

Mean monthly minimum temperatures for Dubbo



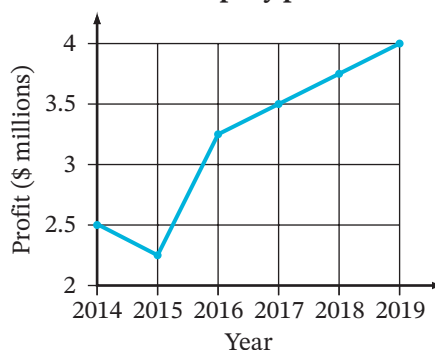
10.01

- Which months have the highest mean minimum temperature? What is that temperature?
- Name any 2 months that have the same mean minimum temperature.
- Which month has a mean minimum temperature of 10°C?
- Between which 2 consecutive months is the smallest drop in mean minimum temperature?

- 5** This line graph shows the yearly profits of a company over a period of 6 years.

- What is incorrect about this graph?
- What misleading impression does this give?
- Redraw the graph correctly.

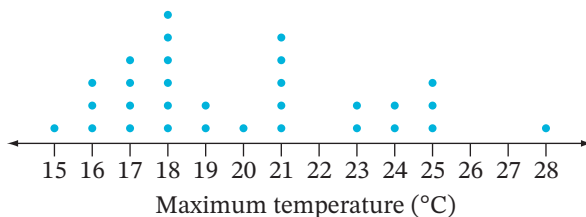
Company profits



10.02

10.03

- 6** The daily maximum temperatures (in $^{\circ}\text{C}$) at Bega during April are shown in this dot plot.



- What is the mode?
- What is the outlier?
- On how many days was the temperature 25°C ?
- On what fraction of days did the temperature drop below 20°C ?

10.04

- 7** This unordered stem-and-leaf plot shows the masses of the players of a football team, in kilograms.

Stem	Leaf
7	9 5 8 1 6 9
8	4 2 5 0 0 8 0 3 2 4 2 8
9	0 8 4 2 2 0 0 8 0 5 6
10	5 4

- How many players are on the team?
- Present this information as an ordered stem-and-leaf plot.
- Which mass occurs most often?
- What is the lowest mass?

For the rest of this exercise, round mean values to 2 decimal places where necessary.

10.05

- 8** For each set of data, find:

i the mean

ii the mode.

a 4 3 2 5 6 4 4

b 6 12 11 12 10 6 6 10 6

c 8 4 1 1 4 1 3 6

10.06

- 9** For each set of data in question 8, find:

i the median

ii the range.

10.07

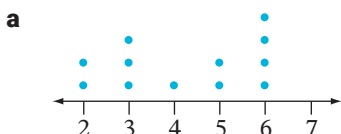
- 10** For each plot below, find:

i the range

ii the median

iii the mode

iv the mean.



b

Stem	Leaf
4	0 0 1 2
5	1 3 4
6	6 6 6 7 8
7	4 5

- 11** The assignment marks for 20 girls and 20 boys are as follows:

Girls:	75	28	37	35	60	73	69	52	94	66
	55	39	48	51	53	18	29	76	59	83
Boys:	88	29	38	72	50	74	73	30	85	10
	28	93	66	17	75	40	55	62	73	58

- Construct a back-to-back ordered stem-and-leaf plot for the data.
- What was the highest mark? Who scored it, a boy or a girl?
- Find the mean, median, mode and range for each group.
- Comment on the differences between the girls and the boys on this assignment.
- Which group of students seemed to be more consistent in the marks they scored, the girls or the boys? Give reasons for your answer.